Fiscal Year:	FY 2020	**Task Last Updated:**	FY 03/26/2020
PI Name:	Rutkove, Seward M.D.		
Project Title:	Musculoskeletal Response to a Partial-gravity Analog in Rats: Structural, Functional, and Molecular Alterations		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	srutkove@bidmc.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-667-8130
Organization Name:	Beth Israel Deaconess Medical Center, Inc./Harvard Medical School		
PI Address 1:	330 Brookline Ave TCC-810		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Artificial Gravity. Appendix D: NASA HRP Artificial Gravity Opportunity
Start Date:	06/09/2016	**End Date:**	01/01/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 1/1/2020 per NSSC information (Ed., 9/17/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 3/11/19) NOTE: Extended to 5/31/2019 per NSSC information (Ed., 10/23/18) NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 12/6/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center/Harvard Medical Center)		
Grant/Contract No.:	NNX16AL36G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Health of the musculoskeletal system depends on exposure to mechanical loading. Removal or reduction of mechanical loading precipitates muscle atrophy and bone loss. Thus, despite rigorous exercise protocols and other countermeasures, muscle atrophy and bone loss are still virtually inevitable consequences of spaceflight. One approach for ameliorating the development of these untoward consequences of space travel is by the induction of artificial gravity during spaceflight via centrifugation. The strength of the artificial gravity depends both on the speed of rotation and diameter of the ship. While it is possible to achieve centrifugal force equivalent 1.0 G, it is also possible that lower levels of force may also be effective for preventing muscle atrophy and irreversible bone loss. In this study, we propose to evaluate the potential effectiveness of artificial gravity for the prevention of muscle atrophy and weakness and bone loss by studying rats using an innovative system for partial unloading that has been developed previously for mice by members of our team. We will study the musculoskeletal response of adult male rats (n=14 / group) following exposure to different degrees of unloading representing 0.2 G, 0.4 G, and 0.7 G as well as fully loaded animals for 1, 2, and 4 weeks. We will study the impact of this partial gravity analog environment on muscle and bone structure and function, including performing histological analyses, in situ force measurement of muscle contraction, as well as measurement of bone mass, microarchitecture, and strength. In addition, we will perform both serological and tissue molecular analyses to better understand the mechanisms and time course of development of these changes at different degrees of unloading. This work will also allow us to more effectively understand the relationship between the development of muscle atrophy and weakness and bone deterioration at varying levels of artificial gravity. At the conclusion of this study, we will have defined the potential impact of partial artificial gravity for prevention of musculoskeletal dysfunction and deterioration. This work will inform the future application of artificial gravity in human research and space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We believe that this research will positively impact in life on Earth since it will inform us about the effects of reduced muscle and bone use that occur normally in our daily lives and as a consequence of disease or injury. For example, using a cane may reduce the normal weightbearing of a single limb. While this obviously may be necessary from an assistive standpoint, this could lead to additional atrophy of the limb muscles and osteopenia of the bone.
Task Progress:	Research accomplishments during this final period: During this final budget period, we completed our first studies using the partial weightbearing system in female animals with good success and recently had a manuscript accepted for publication in Frontiers of Physiology. We also completed our extension of the original grant, in which we sought to evaluate questions of stress associated with our new model, blood flow concerns with use of the pelvic harness for unloading, and evaluation of relative forces placed on the fore and hind limbs of the animals. We are also completing and submitting bone-related manuscripts. Awards, Publications, Presentations: Publications: 1. Mortreux M, Ko FC, Riveros D, Bouxsein ML, Rutkove SB. Longitudinal time course of muscle impairments during partial weight-bearing in rats. NPJ Microgravity 2019 Aug 22;5:20. 2. Mortreux M, Riveros D, Semple C, Bouxsein ML, Rutkove SB. The partial weight-bearing rat model using a pelvic harness does not impact stress or hindlimb blood flow. Acta Astronaut 2020 Mar (168):249-55. 3. Semple C, Riveros D, Nagy JA, Rutkove SB, Mortreux M. Partial weight-bearing in female rats: proof of concept in a Martian-gravity analog. Front Physiol, In Press. Presentations: 1. Mortreux M, Riveros D, Bouxsein ML, Rutkove SB. A moderate daily dose of resveratrol mitigates muscle deconditioning in rats exposed to a Martian gravity analogue. ASGSR, Nov, 2019. 2. Mortreux M, Riveros D, Bouxsein ML, Rutkove SB. A moderate daily dose of Resveratrol mitigates muscle deconditioning in rats exposed to a Martian gravity analogue. Space Health Innovation Conference Nov 2019, San Francisco, CA. 3. Mortreux M, Riveros D, Semple C, Bouxsein ML, Rutkove SB. The partial weight-bearing rat model using a pelvic harness does not induce stress or reduce hindlimb or tail blood flow. Presented at the HRP Investigators Workshop: Galveston, TX, January 2020. 4. Semple C, Riveros D, Nagy JA, Rutkove SB, Mortreux M. Partial weight-bearing in female rats: validation in a Martian-gravity analog. Presented at the HRP Investigators Workshop: Galveston, TX, January 2020. 5. Mortreux M, Riveros D, Semple C, Nagy JA, Bouxsein ML, Rutkove SB. Electrical Impedance Myography to detect muscle disuse in a rat analog model of partial gravity. Presented at the HRP Investigators Workshop: Galveston, TX, January 2020. New technology: None Future research plan: Our work now continues in multiple related directions. We recently had funded or are participating in these two applications: NASA 80NSSC19K1598 (PI: Rutkove) 10/1/2019-9/30/2022. Approaching gravity as a continuum: musculoskeletal effects of fractional reloading. Goal: Study of prevention of reloading injury by providing increasing levels of gravity in a rat model NASA 80NSSC19K1599 (PI: Delp, Co-I: Rutkove) 1/15/2020-1/14/2023. Effects of Simulated Microgravity and Partial Unloading on Organ Systems of the Body. Goal: Determine the influence head-ward fluid shifts play in inducing tissue adaptations with simulated weightlessness. We are also applying for additional grant applications using this partial weightbearing model, including 3 separate application in the most recent Space Biology Solicitation (submitted earlier this month).

Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Mortreux M, Riveros D, Semple C, Nagy JA, Bouxsein ML, Rutkove SB. "Electrical Impedance Myography to detect muscle disuse in a rat analog model of partial gravity." Presented at the 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Small Steps Lead to Giant Leaps: Translating Research into Space Exploration. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Articles in Peer-reviewed Journals	Ko FC, Mortreux M, Riveros D, Nagy JA, Rutkove SB, Bouxsein ML. "Dose-dependent skeletal deficits due to varied reductions in mechanical loading in rats." npj Microgravity. 2020 May 18;6(1):15. https://doi.org/10.1038/s41526-020-0105-0 ; PMID: 32435691; PMCID: PMC7235020 , May-2020
Articles in Peer-reviewed Journals	Semple C, Riveros D, Sung D-M, Nagy JA, Rutkove SB, Mortreux M. "Using electrical impedance myography as a biomarker of muscle deconditioning in rats exposed to micro- and partial-gravity analogs." Front Physiol. 2020 Sep 15;11:1181. https://doi.org/10.3389/fphys.2020.557796 , Sep-2020
Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391 , Apr-2021
Articles in Peer-reviewed Journals	Swain P, Mortreux M, Laws JM, Kyriacou H, De Martino E, Winnard A, Caplan N. "Skeletal muscle deconditioning during partial weight-bearing in rodents – A systematic review and meta-analysis." Life Sci Space Res. 2022 Aug;34:68-86. Review. https://doi.org/10.1016/j.lssr.2022.06.007 ; PMID: 35940691 , Aug-2022
Articles in Peer-reviewed Journals	Swain P, Mortreux M, Laws JM, Kyriacou H, De Martino E, Winnard A, Caplan N. "Bone deconditioning during partial weight-bearing in rodents – A systematic review and meta-analysis." Life Sci Space Res. 2022 Aug;34:87-103. Review. https://doi.org/10.1016/j.lssr.2022.07.003 ; PMID: 35940692 , Aug-2022
Articles in Peer-reviewed Journals	Mortreux M, Ko FC, Riveros D, Bouxsein ML, Rutkove SB. "Longitudinal time course of muscle impairments during partial weight-bearing in rats." npj Microgravity. 2019 Aug 22;5(1):20. https://doi.org/10.1038/s41526-019-0080-5 ; PubMed PMID: 31453318; PubMed Central PMCID: PMC6706399 , Aug-2019
Articles in Peer-reviewed Journals	Mortreux M, Riveros D, Semple C, Bouxsein ML, Rutkove SB. "The partial weight-bearing rat model using a pelvic harness does not impact stress or hindlimb blood flow." Acta Astronaut. 2020 Mar 1;168:249-55. Epub 2019 Dec 24. https://doi.org/10.1016/j.actaastro.2019.12.024 , Mar-2020
Articles in Peer-reviewed Journals	Semple C, Riveros D, Nagy JA, Rutkove SB, Mortreux M. "Partial weight-bearing in female rats: proof of concept in a Martian-gravity analog." Frontiers in Physiology 2020 Apr 3;11:302. https://doi.org/10.3389/fphys.2020.00302 ; PMID: 32308630 ; PMCID: PMC7145975. [Erratum in: Front Physiol. 2020 Jul 03;11:672. PMID: 32719610] , Apr-2020
Significant Media Coverage	Mortreux M, Rutkove S. "Results of resveratrol study showing preservation of muscle mass and function in Martian gravity." Multiple Newspapers and online media. July 2019., Jul-2019